FRI Natural language processing course 2026
Project report

University of Ljubljana
Faculty of Computer and Information Science

AI-Assisted Chatbot for Real-Time Information on Finnish Railway Systems

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Abstract

Keywords
Keyword1, Keyword2, Keyword3 ...

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Introduction

Motivation:

- increasing demand for efficient and personalized customer support [1]
- Traditional methods of obtaining information: waiting in long queues or navigating complex websites, can be time-consuming and inconvenient.
- offering 24/7 availability, quick response times, and the ability to handle multiple queries simultaneously [1]

Features:

- give departure and arrival times of trains (based on train identifier or start and end station + time)
- give travel routes from one station to another
- give delay time of specific trains
- give prices for tickets

Potential datasets:

- Digitraffic Open API (real-time, main source)
- Fintraffic Train Departures: https://junalahdot.fintraffic.fi/?lang=3 (real-time)
- Network Statement of Finnish Transport Infrastructure Agency: PDF file with detailed information on all tracks and stations

Related Work

- Sharma et al. [2]
- Xu et al. [3]
- Rafee et al. [4]: bus enquiry chatbot equipped with voice support and emergency assistance / Features: obtain real-time information related to bus services, including arrival times, route details, and fare estimates / Design: Voice and Text Input Acquisition + NLP Processing and Intent Recognition + Emergency Assistance Information Module + Transport Data Integration+ Voice Response Generation
- Dharani et al. [5]
- Fantin et al. [6]
- Banu et al. [1] : Railway Enquiry Chatbot / Features: ticket availability checks, train schedule inquiries, platform details, and fare estimations / Dataset: train schedules, station details, ticket prices, PNR statuses, live updates on train delays –¿ from public APIs or third-party services, dataset of user queries/ Design: Data Collection Module + Data Processing Module + NLP Module + Database Management + Artificial Neural Network Module + Response Generation
- BetterBahn by Lukas Weihrauch (https://github.com/BetterBahn/betterbahn?tab=readme-ov-file)
- Wang et al. [7]

Testing

Tests with AI-Tool:

Query: ”Hello, i want to go from Helsinki to Turku on Monday 4.5. in the morning. Which trains can I take?”

ChatGPT: Answer: Morning trains (approximate departures) 05:25–05:30 → arrival 07:30 06:50–07:00 → arrival 08:30–08:40 08:30–08:40 → arrival 10:35–10:40 10:30–10:40 → arrival 12:35–12:40

→ nearly all trains correct, but second one doesn't exist → also a concrete service: InterCity train IC 41 departs 06:53 from Helsinki arrives 08:34 in Turku → this is not a train from the official website

Gemini: Gives correct timetable, but one train in between is missing

Follow up question: ”I want to take the one at 08:30, can you tell me where exactly it stops and how much is a ticket?” **ChatGPT:** → ChatGPT gives correct departure & arrival times and also the correct stops in between You also get a information about the ticket price from two different sources, but just a span and not the exact prices

Gemini: → correct departure & arrival times and correct stops → price range and also states additional fees for different classes or bicycle

Follow up question: ”When i want to bring a bicycle, do I have to pay extra?” → ChatGPT states correct information about regulations of taking a bicycle & Gemini already gave this information in the answer before

→ ChatGPT & Gemini can answer question regarding train connections, prices and regulations pretty precisely because it uses Online-Search

Usage Scenarios:

- I want to take a train from Helsinki to Tampere tomorrow around 8 am. What are my options?
- Is my train IC 93 from 8:00 to 9:53 today on schedule?
- How do I get from Tampere to Rovaniemi? (very long, indirect route)
- I want to take train IC 93 tomorrow from 8:00 to 9:53. How much is the ticket?
- I want to take a train from Helsinki to Tampere tomorrow at a flexible time. What's my cheapest option?
- Which cities can I reach from Kuopio without switching trains?
- Which services does my train IC 93 at 8am today from Helsinki provide?
- Was my train on schedule two months ago?
- What trains are arriving to Oulo in the next hour?

Future directions & ideas

By now the following was implemented:

- complete Pipeline: Query → intent extraction → RAG / API → build prompt → LLM answer generator
- API Request handler that can get all informations about train schedules and delays (works)
- RAG handler that scrapes websites with embedding and a faiss index (doesn't work yet on arnes)

What we will work on:

- integrate RAG into pipeline and test it on the cluster
- try different models and compare results
- try different prompt mechanism (more friendly, more fact-based)

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